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# Randomised controlled trial of cardiotocography versus umbilical artery Doppler in the management of small for gestational age fetuses

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**Objective** To compare the impact on use of resources in the management of small for gestational age babies using Doppler ultrasound *versus* cardiotocography.

**Design** A randomised controlled trial.

**Setting** A large district general hospital delivering 5500 to 6000 infants each year, 30% to 35% of which are to women of Pakistani origin.

**Population** One hundred and fifty women delivered of small for gestational age infants.

**Main outcome measures** Primary outcome measures were length of hospital inpatient stay and induction of labour rates. Secondary outcome measures included caesarean section rates and length of stay on neonatal unit.

**Results** The use of Doppler reduced average hospital inpatient stay from 2·5 days to 1·1 days, compared with cardiotocography ($P = 0{\cdot}036$). There was no effect on induction of labour rates or caesarean section rates. There was no significant difference in length of stay on the neonatal unit ($P = 0{\cdot}33$). There was a reduction in monitoring frequency and fewer hospital antenatal clinic visits.

**Conclusion** The use of Doppler ultrasound to manage small for gestational age infants reduces the use of resources, compared with cardiotocography.

## INTRODUCTION

The small for gestational age fetus is at increased risk of perinatal death. In most units the standard management is regular clinical assessment of the pregnant woman with cardiotocography (CTG), either as an outpatient or inpatient. The advent of umbilical artery Doppler examination has been shown to predict[1,2] and prevent[3–5] fetal compromise. The converse also has been shown to be true, and a normal Doppler waveform in a small baby is a good predictor of good fetal outcome[6]. Because Doppler allows doctors to discriminate the small babies most at risk thus allowing closer surveillance, this trial was conducted to determine whether it can also reduce intervention in infants least at risk. If Doppler is superior to CTG, it has the potential to reduce the length of hospital inpatient stay and induction of labour rates since in that event a larger number of small babies will not require intervention.

A previous randomised trial[7] showed reductions in monitoring frequency, antenatal admission and rates of induction of labour and caesarean section for fetal distress. There was also a trend towards fewer admissions to the special care unit. We have repeated this study in a UK district general hospital to assess whether similar results could be obtained.

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## METHODS

The department of obstetrics in Bradford delivers between 5500 and 6000 infants each year, and the proportion of deliveries to women of Pakistani origin is 30% to 35%. At the time of the study deliveries occurred in two hospitals; ultrasound examination was performed at 18 to 20 weeks to confirm gestational age on fetal biparietal diameter and femur length. Further ultrasound examination was performed on the basis of clinical suspicion of an obstetric problem.

This study was conducted to determine differences in management of women with small for gestational age fetuses that would occur, comparing two methods of antenatal clinical assessment. Therefore, patients were randomised either to surveillance by CTG or umbilical artery Doppler examination. The primary trial hypotheses were that the availability of Doppler would reduce antenatal inpatient stay and reduce induction of labour rates. In order to have an 80% chance at the 5% level of showing a reduction in hospital stay of 50%, assuming an average seven day stay with five day standard deviation, we would have required 25 patients in each group. However, in order to show a reduction in induction of labour from 40% to 25% with an 80% chance at the 5% level, 75 patients would be required in each group. On this basis we decided on a sample size of 150 patients.

The entry criterion was a singleton fetus with an ultrasound examination showing the abdominal circumference to be more than two standard deviations below the mean for gestational age on charts recommended by the British Medical Ultrasound Society[8]. There was no gestational age constraint although no patient was entered whose pregnancy was less than 26 weeks. The sole exclusion criterion was fetal malformation diagnosed by ultrasound.

Randomisation was by stratified block randomisation. The four groups consisted of Caucasian primiparous and multiparous women, and Asian primiparous and multiparous women. They were randomised to either Doppler or CTG in blocks of eight using a table of random numbers. The allocated group was placed into sequentially numbered sealed opaque envelopes before commencing the trial. The envelopes were then stored with the Audit Officer at St Luke's Hospital and randomisation was only possible by telephone. The envelopes were cross checked to ensure correct group allocation at the end of the trial.

The Doppler group had umbilical artery Doppler performed using a 4 MHz Doppler probe attached to a Doptech 2001 ultrasound unit with on line waveform analysis. A 100 Hz high pass filter was used to remove background signals. Doppler signals were obtained from the cord without imaging in most situations. Recordings were performed during periods of fetal quiescence and were evaluated over five consecutive heart cycles. All Doppler examinations were performed by J.H. In all cases the A:B ratio was provided for the clinician managing the care. Antenatal CTG was performed as a nonstress test using standard equipment (Hewlett Packard 8041A). Women in both groups were under the care of the same clinicians.

The Doppler group was monitored with a single Doppler examination performed at entry to the trial and then each week for the first three weeks. If normal growth velocity was confirmed on subsequent ultrasound examination, Doppler was performed every two weeks. If there was any concern about the A:B ratio or growth rate, examinations continued each week. Antenatal CTG was not performed on the Doppler group unless there was some other clinical reason, such as presentation with diminished fetal movements or admission for induction of labour. Women in the CTG group were managed as deemed appropriate by their individual clinicians though this usually meant alternate day CTG on an outpatient basis unless other causes for concern developed. There was no routine access to Doppler for the CTG group. Intrauterine growth was assessed every two weeks by ultrasound fetometry, and amniotic fluid volume was semi-quantitatively assessed by subjective impression at the same time. During labour women in both groups were monitored by continuous electronic fetal monitoring. Induction of labour was planned at term + 12 days (as in the general population) unless there was some other factor than the fetus being small for gestational age.

All women in the study, which was approved by the local ethics committee, gave written informed consent. Liaison workers were available to translate for nonEnglish-speaking women.

The primary endpoints were duration of hospital antenatal admission, established from the daily records of admission to the antenatal ward, and induction of labour rates. The reasons for admission, usually a nonreassuring CTG or Doppler particularly with reduced movements, were at a clinician's discretion. Admissions were counted for all reasons including induction or spurious labour. However, although the total duration of all admissions was recorded, the reason for admission was not recorded.

Secondary endpoints were number of investigations (CTG or Doppler), number of outpatient visits to hospital, emergency caesarean section rate, length of stay on the neonatal intensive care unit, birthweight, and 1 min and 5 min Apgar scores.

Data were analysed with the $\chi^2$ test, Mann-Whitney *U* test or Student's *t* tests with unpaired observations as appropriate. *P* values < 0·05 of the outcome variables were regarded as significant. Of the 150 women who took part, 77 were allocated to the CTG group and 73 to the Doppler group. There were no exclusions from the trial, and data were analysed on an intention-to-treat basis.

Within two weeks of delivery all women were sent a questionnaire asking their views on the process of their care, including questions about reassurance received, continuity of care and information given. The questionnaires were also written in Urdu for women who did not read English.

## RESULTS

Seventy-seven women were randomised to a management protocol involving cardiotocography and 73 were allocated a management protocol involving Doppler ultrasound. Table 1 provides details of maternal

**Table 3.** Use of hospital resources. IQR = interquartile range.

| | CTG | | Doppler | | |
|---|---|---|---|---|---|
| | Mean | Median [IQR] | Mean | Median [IQR] | *P* |
| Visits to antenatal clinic | 11·1 | 11 [9–13] | 9·1 | 9 [6–12] | 0·0002 |
| Days in hospital | 2·5 | 1 [1–3] | 1·1 | 1 [0–1] | 0·036 |
| Outpatient CTG | 6·4 | 5 [2–10] | 0·5 | 0 [0–1] | < 0·0001 |
| Inpatient CTG | 4·3 | 2 [0–5] | 2·0 | 1 [1–3] | 0·05 |
| Doppler examinations | 0·1 | 0 [0–0] | 3·8 | 3 [2–5] | < 0·0001 |

**Table 1.** Details of maternal characteristics. Values are given as *n* or mean (SD) unless otherwise indicated. IQR = interquartile range.

| | CTG | Doppler | *P* |
|---|---|---|---|
| Maternal weight (kg) | 59·7 (5·4) | 59·3 (5·0) | 0·8 |
| Maternal height (cm) | 1·60 (0·02) | 1·59 (0·04) | 0·6 |
| Gender of fetus (%) | | | |
| Female | 53 | 50 | |
| Male | 47 | 50 | |
| Parity | | | |
| 0 | 26 | 23 | 0·7 |
| 1 | 29 | 31 | |
| 2 | 9 | 13 | |
| 3 | 8 | 3 | |
| 4 | 2 | 2 | |
| 5 | 1 | 1 | |
| Birthdate: median | 11/07/93 | 18/07/93 | 0·9 |
| [IQR] | [12/03/93–08/01/94] | [01/03/93–24/01/94] | |
| Place of delivery | | | |
| Bradford Royal Infirmary | 54 | 46 | 0·5 |
| St Luke's Hospital | 23 | 27 | |
| Ethnicity | | | |
| Caucasian | 29 | 31 | |
| Noncaucasian | 44 | 46 | 0·6 |
| Smoking (> 10/day) | 28 | 21 | 0·3 |
| Alcohol use | 14 | 12 | 0·5 |
| Husband unemployed | 28 | 25 | 0·5 |
| No live-in support at home | 6 | 2 | 0·05 |

characteristics, showing that the groups of women well-matched in characteristics and expected date of birth to exclude time bias. However, more women in the CTG group had no support at home. Management policy had no detectable effect on fetal outcome (Table 2) and there was no difference in modes of delivery between the groups. In the CTG group 18 women were induced and 17 in the Doppler group ($P = 0·54$; odds ratio 0·9; 95% CI 0·7–4·2).

The use of resources was influenced by the allocation (Table 3). Monitoring a small fetus with CTG surveillance required an average of two additional clinic visits, 1·4 extra days as an inpatient, six extra days as an outpatient, and 2·3 extra inpatient CTG records, compared with Doppler scanning. However, management based on CTG saved 3·7 Doppler investigations per woman. There was no difference in use of resources when comparing Caucasian *versus* nonCaucasian, or primigravid *versus* multigravid women.

**Table 2.** Fetal outcome. Values are given as *n*, mean (SD) or median [interquartile range] where appropriate.

| Infant characteristics | CTG | Doppler | *P* |
|---|---|---|---|
| Gestational age (weeks) | 38·8 (1·9) | 39·2 (1·7) | 0·15 |
| Birthweight (kg) | 2·572 (0·485) | 2·629 (0·433) | 0·5 |
| Weight at 7 months (kg) | 7·80 (1·11) | 7·50 (1·03) | 0·15 |
| Apgar score | | | |
| At 1 min | 9 [8–9] | 9 [8–9] | 0·8 |
| At 5 min | 9 (9–9) | 9 (9–9) | 0·8 |
| Stillbirth | 1 | 0 | |
| Neonatal death | 0 | 1 | |

One infant died in each group. The mother of the infant that died in the Doppler group presented at 31 weeks with hypertension and ultrasound demonstrated a small fetus. Immediately after randomisation, when the Doppler probe was attached, the fetal heart rate was 60 beats per minute and persisted at this rate. The woman was delivered by caesarean section within 1 hour of entry to the trial, but the infant was severely asphyxiated and died on the first day of life. The infant that died in the CTG group died *in utero* at 37 weeks and weighed 1900 g. The fetus was detected as small at 35 weeks and CTG records were normal and reactive on three occasions. The last normal CTG record was two days before delivery and fetal movements were normal up to 12 h before a routine CTG record. The woman was booked for a CTG record that morning but the fetal heart was absent; postmortem was declined.

Five of the fetuses monitored by Doppler had an A:B ratio above the 95th centile. All but the neonatal death had a good outcome. The number of nonreassuring CTG records was not recorded.

Seventeen infants whose mothers were allocated CTG monitoring were admitted to special care baby units for a total of 171 days (mean 2·22, SD 6·81; median 0, interquartile range 0–0). Twelve infants whose mothers were monitored by Doppler were admitted to special care units for a total of 97·5 days (mean 1·34, SD 4·21; median 0, interquartile range 0–0). These differences were not statistically significant by Mann-Whitney $U$ test ($P = 0·33$). The morbidity on the special care unit was similar for both groups (OR 2·3; 95% CI 0·7–7·7) (Table 4).

**Table 4.** Neonatal problems on special care baby unit. Values are given as *n*. BM = capillary blood glucose on Boehringer Mannheim BM-Accutest strips.

| | CTG | Doppler |
|---|---|---|
| Jaundice needing phototherapy | 3 | 0 |
| Jittery | 2 | 1 |
| Fits | 1 | 0 |
| Ventilated | 1 | 2 |
| Hypoglycaemia (BM = 0) | 2 | 1 |

There was no significant difference in either elective (4 CTG *vs* 7 Doppler) or emergency (15 CTG *vs* 9 Doppler) caesarean sections in either group: OR 0·5, 95% CI 0·1–1·8 and OR 1·7, 95% CI 0·7–4·2, respectively. There was one infant born iatrogenically before 34 weeks in each group. The number of babies born iatrogenically before 37 weeks was not statistically different between each group (two in the control group, one in the Doppler group).

There were 45/77 questionnaires (58·4%) returned in the control group, and 52/73 (71·2%) returned in the Doppler group. There were no differences between groups in satisfaction with care, the level of reassurance provided by tests or of continuity of care. The women expressed no preference regarding type of care they would prefer in a future pregnancy if the fetus was small for gestational age.

## DISCUSSION

When a fetus is small for gestational age there is a concern that there may be intrauterine growth retardation. This leads to closer surveillance of small fetuses to try and detect early hypoxaemia. This would ideally allow intervention when appropriate and avoid intervention in small but healthy fetuses.

Cardiotocography has been the standard clinical tool used to monitor at risk pregnancies despite a lack of evidence from randomised controlled trials to support its use[9]. Randomised trials that are available suggest outpatient use of CTG increases morbidity[9] and mortality. Doppler umbilical artery velocimetry has been shown in randomised controlled trials to improve perinatal outcome in at risk pregnancies, both in terms of perinatal death and morbidity[3–5].

In our study the aim was to determine whether the use of Doppler would reduce hospital intervention, both in terms of admission and rates of induction, with similar clinical outcome. A similar study had been performed in Sweden and this showed a reduction in hospital admission and rates of induction of labour and emergency caesarean section for fetal distress. There was also a trend towards reduced neonatal intensive care admission[7]. Our trial was a pragmatic trial—there was not a rigid protocol except that clinicians usually felt that a CTG record gave reassurance for 48 h to 72 h and a Doppler examination for a week or more.

The rationale behind using Doppler as an alternative to CTG is that it has been shown to give an earlier warning of impending asphyxia[10]. There is also good evidence that the presence of a normal umbilical artery waveform is associated with a good outcome[6]. This should, therefore, allow better targeting of intervention.

The other advantage of umbilical artery Doppler is that it can be done more quickly than a CTG record and requires less frequent examinations. This allows fewer hospital attendances. It is also more reproducible and less open to inter-observer variation so more certainty can be gained from a single result[11,12]. This would lead to less hospital admissions for further surveillance. The advent of computerised CTG monitors may allow a re-emergence of cardiotocography as a method of fetal surveillance, although this has not yet been subjected to randomised controlled trials.

In our study we did show a reduction in hospital admission using Doppler from an average of 2·5 days to an average of 1·1 days per woman. There were also less monitoring occasions and fewer hospital antenatal clinic visits. However, there was no difference in induction of labour rates or modes of delivery. There was no significant difference in any other outcome measure or in any other intervention.

Prior to this study Doppler had been used only in very highly selected pregnancies in Bradford. It was an extension of Doppler in the unit to use it in all small for gestational age fetuses. It may well be that when clinicians are more confident with the technique, operational differences when compared with CTG may be even greater. Since the end of the trial we now find that we rarely admit women with a small for gestational age fetus. The fact that we have a high Asian population means we see more women whose fetuses are small, which fact may be constitutional rather than pathological. However, it is not possible to ignore a small for gestational age fetus whatever its mother's ethnicity. As yet it is not certain as to whether small Asian fetuses do better than small Caucasian fetuses, therefore, pragmatically all small fetuses need monitoring.

The power calculations that were made to establish sample size were based on data obtained before the opening of our day assessment unit. The opening of this facility obviously had an effect on lengths of stay in hospital in both groups. Day care has been shown to reduce induction of labour rates also, and it may have been that this was responsible for the reduction in induction of labour rates rather than any effect of either Doppler or CTG[13]. This difference between estimated and actual lengths of stay reduces the power of the study. The fact that the power of the study was diminished means that it is less able to exclude a difference between the two study groups which is small and not detected because of the sample size.

The duration of hospital stay was recorded from the admission registers of the antenatal ward. It was not possible to determine whether admission was related to the presence of a small baby or an abnormal monitoring episode. We felt that trying to ascribe a reason for admission would increase the subjectivity and allow possibility of bias in terms of comparing lengths of stay. We felt that if all episodes were included then the admissions for reasons other than abnormal monitoring episodes should be the same in each group. Hence, the difference between the two groups should reflect the method of monitoring. It was felt that this was more likely to be a 'hard endpoint'. The fact that Dopplers can be undertaken each week and CTG records obtained every other day inevitably leads to more monitoring with the use of CTG records. The aim of this study was not only to establish that a difference did occur but to give some estimate of the size of that difference. One of the great difficulties with CTG is its subjective nature. This means that there are more likely to be false positive results which will lead to an increase in monitoring and admissions for reassurance of both the clinician and the patient. This also made it impossible to determine with certainty when CTG records were nonreassuring. It is, therefore, one of the benefits of Doppler that the monitoring frequency required for reassurance is less. Also the fact that Doppler has less false positive results leads to less admissions for further assessment. The false negative result with CTG leading to stillbirth is a concern. However, CTG has not been shown to be a good predictor of outcome[9]. This occurrence highlights the lack of reliability of CTG when used to monitor at risk pregnancies on an outpatient basis.

Doppler ultrasound seems to be at least as safe as CTG monitors of small fetuses and, with the support of previous randomised controlled trials, may be safer. It also leads to less intervention in terms of hospital admission without an increase in iatrogenic premature delivery. We would suggest, in an otherwise uncomplicated pregnancy, that Doppler examination alone, rather than CTG, be used to monitor a small fetus on an outpatient basis.

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